Calcitriol Treatment Is Not Effective in Postmenopausal Osteoporosis

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Study Objective: To determine if calcitriol is an effective treatment in postmenopausal osteoporosis.

Design: Double-blind, randomized clinical trial of 2 years' duration.

Setting: University medical center with patients recruited by media announcements.

Patients: Eighty-six postmenopausal women with vertebral compression fractures.

Interventions: Patients were treated with calcitriol or placebo. Mean dose was 0.43 μg/d. Dietary calcium was 1000 mg/d (24.9 mmol/d). The medication dose and dietary calcium were adjusted for hypercalciuria or hypercalcemia.

Measurements and Main Results: No significant differences between placebo and control groups were seen in the percent change in total body calcium (0.4% ± 1.0 compared with 0.0% ± 0.9), single photon absorptiometry (−0.5% ± 1.2 compared with −3.1% ± 0.9) or dual photon absorptiometry (0.0% ± 1.7 compared with −1.0% ± 2.2). New fractures were seen in 16% of the placebo group and 26% of the calcitriol group, so the difference in percent fractures was 10% (95% CI, −5.7% to 25.7%). Bone biopsies did not show changes in either group. The calcitriol group had significantly higher serum and urine calcium values, but renal function was not worse than in the placebo group.

Conclusions: Calcitriol is not an effective treatment for established postmenopausal osteoporosis.

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Calcitriol (1,25-dihydroxy-vitamin D) is an important hormone involved in the regulation of calcium metabolism. Although it is well known that calcitriol increases the gastrointestinal absorption of calcium, its actions on bone are not fully understood (1). In vitro studies suggest that it increases bone resorption (2), and it may act to differentiate bone cell precursors (1). Calcitriol also may act on the parathyroid gland, modifying the point at which serum calcium levels inhibit parathyroid hormone release (3).

The role of calcitriol in the pathogenesis of postmenopausal osteoporosis is also uncertain. Studies on the serum levels of calcitriol in postmenopausal women have shown conflicting results, which is not surprising in view of the wide normal range and the difficulties in measuring this steroid (4, 5). There is evidence that calcitriol deficiency may be involved in at least some types of postmenopausal osteoporosis. Low gastrointestinal absorption of calcium has been documented in postmenopausal women, and this absorption can be increased with calcitriol (6, 7). Although basal calcitriol levels may be normal in postmenopausal women, the levels do not rise as expected after an infusion of parathyroid hormone (8, 9). Thus, Riggs and Melton (10) have postulated that one type of osteoporosis is caused by a primary failure of the kidney to produce sufficient calcitriol, with subsequent low absorption of calcium and negative calcium balance.

On the other hand, calcitriol deficiency may not be a pathogenic factor for other types of postmenopausal osteoporosis. Most studies of bone mass show a marked decrease at the time of menopause, and this loss of bone mass may manifest itself as osteoporosis in later years. A recent study (11) showed that the bone loss in normal women followed longitudinally through menopause is not associated with any change in vitamin D levels. Women who are perimenopausal show a normal rise in calcitriol levels after an infusion of parathyroid hormone (12). It is also possible that women who do have low calcitriol levels may have a secondary suppression in calcitriol synthesis, caused by a primary increase in bone resorption with excess mineral released from bone (10). Therefore, it is difficult to predict whether exogenous calcitriol would benefit women with osteoporosis. It could cause beneficial effects by enhancing calcium absorption or acting directly on bone to increase formation (13, 14). However, it could also have harmful effects by increasing bone resorption.

Several clinical trials had previously been conducted to examine the in vivo effects of calcitriol or its analog, 1-alpha-hydroxy-vitamin D, on women with osteoporosis (4, 15-25). These studies yielded conflicting results, with bone mass or fracture rate improving (15, 19, 21, 24, 25), worsening (17, 20, 23), or remaining unchanged (16, 18, 22). Results on fracture rates have also been conflicting. Therefore, we performed a double-blind, randomized clinical trial for 2 years. The primary purpose of the study was to determine whether calcitriol could increase bone mass. We also sought to evaluate side effects of the drug, particularly on the kidney. Finally, we wanted to see if calcitriol had any effects on bone remodeling in women with postmenopausal osteoporosis. Such information could help us understand the pathophysiology of this common disease.

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Methods

Patients

Eighty-six women recruited by media advertisements were enrolled. All were postmenopausal, between ages 50 and 80. Lateral radiographs of the spine were examined, and women were included if they had at least two compression fractures (> 15% reduction in anterior height) without history of serious trauma. All participants were white, were taking no medications for treatment of osteoporosis (except calcium supplements in some cases), and were ambulatory. All had normal results from tests for serum electrolytes, calcium, phosphate, alkaline phosphatase, liver function, creatinine, thyroid function, protein electrophoresis, urinalysis, hematocrit, and leukocyte count. Women with a history of corticosteroid use, malnutrition, sarcoidosis, liver disease, rheumatoid arthritis, nephrolithiasis, renal disease, or recent malignancy were excluded. Each woman gave informed consent as approved by the Institutional Human Subjects Review Committee.

Study Design

All women were initially placed on a calcium intake of 24.9 mmol/d (1000 mg/d), using supplements if necessary. Upon enrollment each woman received a bottle of capsules supplied by Hoffmann-La Roche (Nutley, New Jersey). The bottles had been randomized and contained either 0.25 μg calcitriol or an identical placebo. The women started taking one capsule twice a day. At 2-week intervals the dose was increased by 1 or 2 capsules to a maximum of 8 capsules daily. Serum calcium, creatinine, and phosphate levels were measured every 2 weeks for the first 3 months, then every 6 weeks. These samples were taken while the patient was fasting or after eating a zero-calcium breakfast. Urinary calcium (24 hour) was measured at 4 and 12 weeks, then every 12 weeks. Every 6 months the patients ingested a 2-day hydroxyproline-free diet and urinary hydroxyproline (24 hour) was subsequently measured. If the serum calcium level was greater than 2.54 mmol/L (10.2 mg/dL) or the urinary calcium greater than 8.75 mmol/d (350 mg/L), then the dose was not increased and the calcium intake was reduced. If the serum or urinary calcium remained high, then the dose was decreased. After the first 12 weeks, most patients were stable, but in some cases hypercalciuria subsequently developed, which required further dose reductions.

The study duration was 2 years. Serum and urine chemical measurements were made every 6 to 12 weeks. Bone mass was measured every 3 to 6 months. Radiographs of the lateral spine were done yearly. Bone biopsies were done at the beginning and end of the study. Formal assessment of calcium intake was done at baseline, 36 weeks, and 2 years by a nutritionist; patients were also questioned about their calcium intake at least every 3 months. A physical examination and electrocardiogram were done yearly, and blood pressure was recorded every 6 weeks.

Serum and Urine Tests

Tests for serum calcium, phosphate, alkaline phosphatase, electrolytes, uric acid and creatinine, and urine calcium and creatinine were done using an auto-analyzer in the clinical laboratory of the University Hospital. At 6-month intervals tests for hematocrit, leukocyte count, and aspartate aminotransferase were done with standard automated methods. Total urinary hydroxyproline was assayed according to a reported method (26), except decolorization was added and the pH was adjusted to exactly 6.0. The Hoffmann-La Roche laboratory measured the vitamin D levels and the parathyroid hormone levels, using an assay for the midportion of the molecule.

Bone Mass Measurements

The methods for measuring bone mass have previously been reported (27). Total body calcium by neutron activation analysis was measured every 6 months. Using this technique, the subject is exposed to a neutron beam that converts calcium-48 to calcium-49, which is then quantitated in a whole body counter along with calcium standards that were exposed simultaneously. Because over 99% of the calcium in the body is stored in the skeleton, this represents a measure of total skeletal mass.

Single photon absorptiometry of the distal radius was performed every 3 months at a site 10% the length of the forearm proximal to the styloid process. Dual photon absorptiometry of the lumbar spine (L1-4) was measured every 6 months. We quantitated the grams per vertebra and divided the total by the height of the vertebral bodies, resulting in units of g/cm. Quantitative computed tomography was performed yearly on those patients who agreed to the procedure. The central cancellous bone of the spine was measured from T12 to L3.

Radiographs

Standard radiographs were taken of the lateral thoracic and lumbar spine. Anterior and posterior heights of each vertebral body from T4 to L4 were measured. The initial fracture index was the cumulative sum of the percent reduction in anterior height for each vertebra that was compressed more than 15%. A new fracture was defined as an anterior height loss of greater than 15% resulting in an anterior/posterior ratio of less than 85%. None of the patients developed new compression of the posterior height of greater than 15%.

Bone Biopsies

Biopsy samples were taken under local anesthesia, using an electric drill (outer diameter 6 mm, inner 4 mm) vertically from the anterior iliac crest. These samples were fixed and embedded in methacrylate. Sections (20 mm^2, using two sections if necessary) which were unstained or stained by Goldner method or with a stain for acid phosphatase (28) were quantitated, using a digitizing pad connected to a computer while examining the slide under a microscope equipped with a drawing arm. Tetracycline deposits in bone and can be used as a label. The women were given a 3-day label of demeclocycline; after 8 to 19 days they received a 3-day label of tetracycline and biopsies were performed a week later. Only cancellous bone was measured, and the transitional zone (within 1 mm of the cortex) was not included. The mineralizing surface was defined as the total length of the demeclocycline label. Osteoid was measured at a thickness greater than 2 μm. The measurements are expressed in the new standardized nomenclature for bone histomorphometry (29). We included bone volume per tissue volume, osteoid surface per bone surface, number of osteoclasts, osteoid thickness, mineral apposition rate, mineralizing surface, bone formation rate per bone surface and per bone volume, and trabecular thickness and number. The latter were calculated using a plate model (30).

Statistical Analysis

The data was stored and analyzed using a database system (Medical Research Time-Oriented Database, Retriever Data Systems, Seattle, Washington). The study was designed primarily to assess change in bone mass. For each woman, linear regression was done of total body calcium, single or dual photon absorptiometry on time in weeks. The predicted values from the regression equation for weeks 0 and 100 were then used to calculate a percent change. The values for the placebo group were then compared to the treated group using the Student *t*-test. For the bone mass measurements and

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Table 1. *Baseline Measurements for Women with Postmenopausal Osteoporosis Who Had Been Randomly Assigned to Receive Placebo or Calcitriol**

Variables	Placebo ($n = 43$)	Calcitriol ($n = 43$)
Age, *y*	67.1 ± 1.2	67.9 ± 1.0
Height, *cm*		
Previous	164 ± 1	164 ± 1
Current	159 ± 1	159 ± 1
Bone mass		
Total body calcium, *g*	665 ± 16	657 ± 19
Single photon absorptiometry, *g/cm*	0.659 ± 0.018	0.652 ± 0.023
Dual photon absorptiometry, *g/cm*	3.87 ± 0.13	3.67 ± 0.14
Fracture index†	41.8 ± 7.0	34.3 ± 6.6
Serum chemistry		
Calcium, *mmol/L*	2.27 ± .02	2.24 ± .01
Phosphate, *mmol/L*	1.26 ± .02	1.26 ± .02
Alkaline phosphatase, *μkat/L*	1.3 ± 0.1	1.2 ± 0.1
Creatinine, *μmol/L*	76 ± 3	79 ± 2
Parathyroid hormone, *ng/ml*	0.43 ± 0.03	0.41 ± 0.03
1,25 $(OH)_2$ Vitamin D, *pg/ml*	24.7 ± 1.8	24.6 ± 1.6
25 (OH) Vitamin D, *ng/ml*	26.3 ± 2.4	26.7 ± 1.9
Urine values		
Calcium, *mmol/d*	3.4 ± 0.3	3.3 ± 0.3
Hydroxyproline, *mg/d*	15.9 ± 0.8	14.7 ± 0.9
Creatinine clearance, *mL/s*	1.08 ± 0.05	1.00 ± 0.05
Calcium intake, *mmol/d*	20.4 ± 0.6	20.6 ± 2.12

* Data are mean ± SE. No significant differences were seen between the groups.
† See Methods section for derivation.

the biochemical data, the values in the calcitriol compared with placebo groups were compared at baseline, 48, and 96 weeks by *t*-tests (or Wilcoxon two group test if the values were not normally distributed). This resulted in multiple *t*-tests, but no correction was made, because the only significant biochemical differences between placebo and calcitriol reflected the expected change in the serum and urine calcium values. The values within each group were also tested using one way analysis of variance for repeated measurements. If a significant difference was seen, then paired *t*-tests were done between the value at baseline and at 48 or 96 weeks; also, the percent difference for placebo and calcitriol was compared. For fracture incidence 95% confidence intervals were calculated (31). Data are shown as mean ± standard error.

Results

Initially there were no differences between the placebo group and the calcitriol group in age, height, bone mass, fracture index, biochemical values, or calcium intake (Table 1). Of the 86 women enrolled in the study, 76 completed at least 1 year and 72 completed 2 years. Four women on calcitriol discontinued the study within 12 weeks for personal reasons. Four others discontinued after a year; 1 for personal reasons and 1 for multiple fractures; 1 was placed on corticosteroids for a pulmonary disease, and 1 developed clinical signs of chronic lymphocytic leukemia. The initial bone biopsy sections from the patient with leukemia were reviewed by a hematologist who felt the disease had been present in a mild form before the study. Six women receiving placebo discontinued within 6 months: 4 for personal reasons, 1 for severe depression, and 1 woman died of a myocardial infarction.

The mean dose of calcitriol in the treated group was 0.76 ± 0.07 μg/d at 12 weeks and 0.43 ± 0.03 μg/d at 96 weeks. The dose of placebo was 1.59 ± 0.09 at 12 weeks and 1.51 ± 0.11 at 96 weeks. The calcium intake was adjusted during the study for hypercalciuria; mean calcium intake for each group is shown in Figure 1. During the study calcium intake was significantly higher in the placebo group than in the calcitriol group. The calcium intake was significantly higher than baseline in the placebo group and lower than baseline in the calcitriol group. Because this was a double-blind study, some of the placebo patients had been placed on a low calcium intake if they developed hypercalciuria, which probably resulted from the increased calcium intake.

Bone Mass Measurements

The mean percent change in total body calcium over 100 weeks, as determined by linear regression, was 0.4% ± 1.0% in the placebo group and 0.0% ± 0.9% in the calcitriol group ($P > 0.3$). For dual photon absorptiometry the values were 0.0% ± 1.7% for placebo and −1.0% ± 2.2% for calcitriol ($P > 0.3$). Single photon absorptiometry values showed −0.5% ± 1.2% for placebo and −3.1% ± 0.9% for calcitriol ($P = 0.09$). For most individual women it was difficult to determine if there had been a gain or loss of bone. The precision of these techniques is 2% to 3% in normal persons (27) so that in an individual a change of 6% would be significant. If at least two methods of bone mass quantitation demonstrated a gain or loss of more than 6% per 100 weeks, then 9 women improved (5 placebo, 4 calcitriol) and 15 worsened (6 placebo, 9 calcitriol).

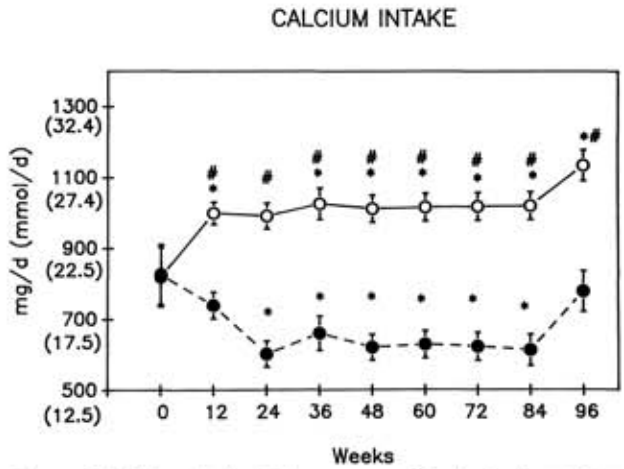


Figure 1. Calcium intake. Data are mean with standard error bars. Number sign = $P < 0.05$, calcitriol compared with placebo. Asterisk = $P < 0.05$, paired *t*-test from baseline.

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Table 2. *Bone Mass Measurements in a Randomized, Double-blind Trial of Calcitriol in Postmenopausal Osteoporosis**

Variable	Placebo		Calcitriol	
Total body calcium by neutron activation, *g*				
Baseline	664 ± 17	(35)†	652 ± 20	(37)
1 y	673 ± 18	(33)	649 ± 21	(33)
2 y	659 ± 20	(28)	648 ± 21	(28)
Single photon absorptiometry, *g/cm*				
Baseline	0.656 ± 0.021	(37)	0.645 ± 0.021	(39)
1 y	0.665 ± 0.021	(36)	0.637 ± 0.022	(38)
2 y	0.649 ± 0.021	(37)	0.629 ± 0.019‡	(33)
Dual photon absorptiometry, *g/cm*				
Baseline	3.85 ± 0.14	(35)	3.66 ± 0.15	(34)
1 y	3.77 ± 0.11	(36)	3.61 ± 0.14	(39)
2 y	3.73 ± 0.12	(36)	3.52 ± 0.13	(33)
Quantitative computed tomography, mg/cm^3				
Baseline	67.8 ± 4.9	(33)	62.4 ± 6.2	(29)
1 y	78.4 ± 5.2	(16)	70.6 ± 5.6	(17)
2 y	73.7 ± 5.2	(13)	76.0 ± 12.9	(7)

* This table includes only women who completed at least 1 year of the study. Data shown are mean ± SE. No differences were seen between placebo and calcitriol groups.
† Numbers of women are shown in parentheses.
‡ $P < 0.01$ by paired *t*-test from baseline.

One-way analysis of variance using repeated measurements did not show any significant differences over time in total body calcium or dual photon absorptiometry in either group. There were no changes in single photon absorptiometry in the placebo group, but in the calcitriol group a significant decrease was seen after 2 years (Table 2). The percent changes for the bone mass measurements are shown in Figure 2; no significant differences were seen between placebo and calcitriol groups at baseline, 48 weeks, or 96 weeks.

A small number of patients underwent repeated measurements using quantitative computed tomography. The percent change after 1 year was 8.0 ± 4.1 in the placebo and 9.8 ± 7.6 in the calcitriol group. After 2 years the values were −6.6 ± 1.9 and −13.4 ± 4.0, respectively ($P = 0.10$, placebo compared with calcitriol). There was no significant change within either group.

Biochemical Measurements

Figure 2 and Table 3 show that serum calcium increased significantly in the calcitriol group and was higher than in the placebo group. Similar changes were seen with urinary calcium. There were no significant differences between calcitriol and placebo groups in other biochemical variables measured, including phosphate, alkaline phosphatase, creatinine, uric acid, urinary hydroxyproline, creatinine clearance, leukocyte count, or hematocrit. The creatinine and uric acid increased slightly in both groups, while hematocrit and phosphate decreased in both groups. There was no correlation between the change in phosphate and the change in serum calcium or calcium intake in either group.

The parathyroid hormone level decreased from 0.43 ± 0.03 to 0.34 ± 0.03 ng/mL in the placebo group ($P = 0.009$) and from 0.41 ± 0.03 to 0.30 ± 0.04 ng/mL in the calcitriol group ($P = 0.05$). There was no difference between the calcitriol and placebo groups in absolute levels of parathyroid hormone or in change of parathyroid hormone. The percent change was not used in this test because many of the values were near the limit of detection by the assay, so small changes led to large percent changes.

Fractures

There were 19 new vertebral compression fractures in nine women in the calcitriol group and 9 fractures in six women in the placebo group. Thus 16% of the placebo group and 26% of the calcitriol group had at least one new fracture. The difference in percent fractures was 10% (CI, −5.7% to 25.7%). There were no hip fractures in any patient during the study. Five women on calcitriol had other fractures (wrist and pelvis, radius, wrist, finger, and ankle) and two women on placebo had other fractures (rib, hand).

Bone Biopsies

No differences were seen between calcitriol and placebo groups at baseline or at study completion (Table 4). The calcitriol treatment did not increase bone formation or improve bone architecture. The number of osteoclasts, an indicator of resorption, did not increase.

Clinical Findings

No major clinical side effects of calcitriol were identified during this study. Although the urinary calcium was markedly increased in some patients (up to 15 mmol/d or 600 mg/d), this increase was not associated with symptoms. These high levels of urinary calcium responded to decreased dose or calcium intake, so no patient sustained a urinary calcium greater than 8.7 mmol/d (350 mg/d). The serum calcium increased in the calcitriol group, but it exceeded the normal range in only eight women. The highest value was 2.79 mmol/L (11.2 mg/dL). The serum calcium returned to normal with adjustments in dose or calcium intake. No changes were seen in blood pressure. Liver function tests, electrolytes, electrocardiogram findings, and physical examinations did not show any consistent changes. Minor symptoms such as upper respiratory or urinary tract infections, back pain, and indigestion were similar in both groups.

Discussion

Treatment with calcitriol for 2 years did not improve the bone mass in women with postmenopausal osteoporosis. No increases were seen in the total skeleton as measured by total body calcium; in the appendicular

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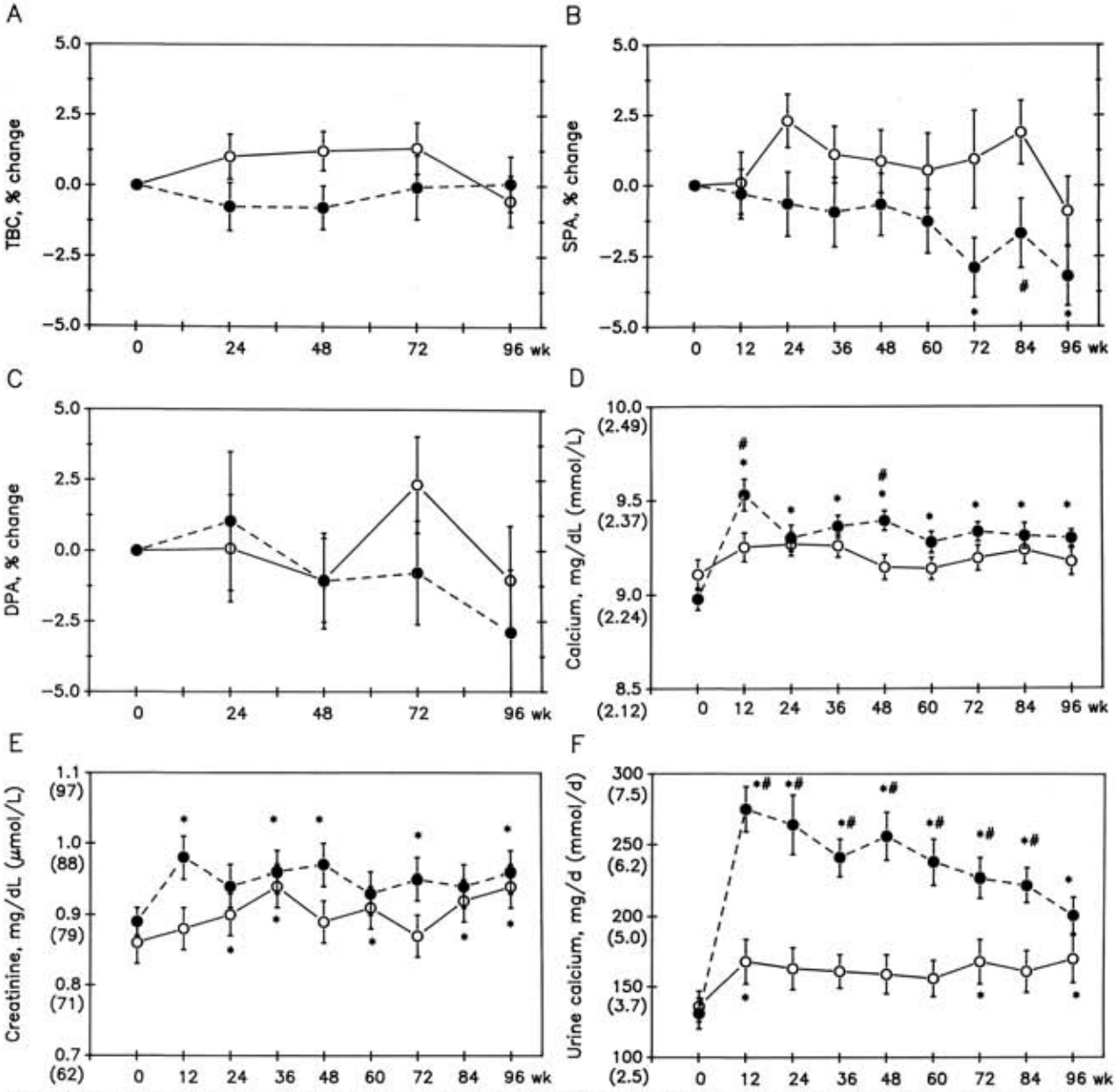


Figure 2. Changes in bone mass measurements, serum and urine calcium, and serum creatinine. Data are mean with standard error bars. The placebo group is shown in open circles with solid lines, and the calcitriol group in closed circles with broken lines. Number sign = $P < 0.05$, calcitriol compared with placebo. Asterisk = $P < 0.05$, paired *t*-test from baseline. Figure 2A indicates the percent change in total body calcium (TBC) by neutron activation. Figure 2B shows the percent change in single photon absorptiometry (SPA) values. Figure 2C indicates the percent change in dual photon absorptiometry (DPA) values. Figure 2D shows the serum calcium levels; Figure 2E, the serum creatinine levels; and Figure 2F, the urine calcium levels.

cortical bone as measured by single photon absorptiometry; or in the cortical and cancellous bone of the vertebrae, as measured by dual photon absorptiometry. The goal of treatment of postmenopausal osteoporosis is to prevent further fractures. However, in our sample there was no trend toward an improvement in fracture rate.

Persons with low bone mass are more likely to sustain a fracture than those with high bone mass, but other factors may also play a role (32). Thus, bone that has discontinuous architecture may be more likely to fracture than bone with similar mass but intact structure (33). The trabecular width and separation, indicators of structure, did not change with calcitriol. Another factor that may lead to fractures is accumulation of microfractures in osteoporotic bone (34, 35). It is possible that bone with a higher formation rate could repair these microfractures, thus improving fracture risk. However, in our study the calcitriol did not increase the bone formation rate. Thus, it seems unlikely that an agent which does not increase bone mass, improve bone architecture, or increase bone formation rate would be able to prevent further osteoporotic fractures.

There are several possible explanations for the stable bone mass in our placebo patients. Normal women show a relatively rapid loss of bone mass immediately after the menopause, losing up to 2% each year (36). After that the rate slows to less than 1% per year (36-38). The mean age for the women in our study

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Table 3. *Percent Changes in Serum and Urine Values in a Randomized, Double-blind Trial of Calcitriol in Women with Postmenopausal Osteoporosis**

Variable	Placebo		Calcitriol	
	1 y	2 y	1 y	2 y
Serum				
Calcium	0.7 ± 0.9	1.0 ± 0.8	4.8 ± 0.7†‡	3.8 ± 0.7†‡
Phosphate	−9.8 ± 2.4†	−5.6 ± 2.7†	−12.3 ± 1.8†	−8.5 ± 2.1†
Alkaline phosphatase	0.5 ± 2.5	4.4 ± 3.1	−4.0 ± 3.5	6.1 ± 5.5
Creatinine	8.2 ± 3.3	14.2 ± 3.6†	9.6 ± 2.3†	9.8 ± 3.5†
Uric acid	9.6 ± 2.6†	12.8 ± 2.8†	10.6 ± 3.9†	19.7 ± 4.1†
Urine				
Calcium	30 ± 13	40 ± 21†	150 ± 25†‡	110 ± 26†‡
Hydroxyproline	7 ± 8	2 ± 7	37 ± 16	29 ± 14
Creatinine clearance	1.4 ± 4	−6.5 ± 4†	0.7 ± 4	−3.0 ± 4
Leukocytes	−7.8 ± 0.8†	−30.8 ± 7.3	−5.3 ± 2.8	−23.5 ± 8.3†
Hematocrit	−2.2 ± 0.8†	−4.6 ± 1.1†	−2.3 ± 1.1	−3.6 ± 1.2†

* Data are mean ± SE.
† = $P < 0.02$ when the values were compared to baseline by paired t-test.
‡ = $P < 0.01$, comparing percent change in placebo compared with calcitriol.

was 68 years, so some of them may have already lost the bone mass and stabilized. In osteoporotic women there may be two stages of disease, an initial stage when bone is lost and a later stage when bone mass is no longer declining but fractures continue due to the low mass. Longitudinal studies have suggested that bone is lost as a proportion of the existing bone (38). When fewer bone surfaces are available, bone cannot be lost as rapidly. Bone biopsies in our patients showed a reduced surface area and little resorption, which would be expected if bone loss were stable.

Finally, the bone mass may have been stable in our placebo patients because we ensured adequate calcium intake. In our center, a past study of osteoporosis with a placebo group not given any calcium supplements showed a 1.3% per year rate of decrease of bone mass (39). In two other studies the control group received calcium supplements without a significant change in total body calcium after 2 years (40, 41). Thus it is possible that the calcium supplements prevented some bone loss. It is difficult to conclusively prove such a protective effect of calcium on bone mass, since a non-calcium control group is rarely available due to suggestions from the media regarding increased calcium intake. It is of interest that the baseline calcium intake of women in this study averaged 823 mg/d (20.5 mmol/d), compared to an average of 480 mg/d (12.0 mmol/d) in a previous study started 10 years ago (40).

The serum phosphate decreased in both groups, without difference between the groups. Although no correlation was seen between change in calcium intake and change in serum phosphate, we did not directly measure calcium absorption, so the increased calcium intake may have contributed to the decrease in serum phosphate. Significant decreases in phosphate were also noted in a previous study (40) in the group treated with placebo and calcium supplements. The decrease in phosphate could also have resulted from the fact that all baseline samples were taken while the women were inpatients at the research center, after an overnight fast, but some of the follow-up samples were taken after a zero-calcium breakfast. The carbohydrates in the breakfast could lower serum phosphate.

Calcitriol has been shown to increase the intestinal absorption of calcium in patients with osteoporosis (6, 7). This factor was not measured directly in our study, but the effect of increased absorption—increased urinary calcium excretion—did occur. Thus,

Table 4. *Bone Histomorphometric Values from Iliac Crest Cancellous Bone in a 2-Year Randomized, Double-blind Trial of Calcitriol in Women with Postmenopausal Osteoporosis**

Variable	Placebo		Calcitriol	
	Initial ($n = 42$)	Final ($n = 32$)	Initial ($n = 41$)	Final ($n = 30$)
Bone volume/tissue volume, %	14.4 ± 0.7	13.9 ± 0.8	13.9 ± 0.8	13.0 ± 0.6
Osteoid surface/bone surface, %	10.5 ± 0.9	8.6 ± 1.0	10.2 ± 1.2	10.3 ± 1.5
Number of osteoclasts/mm^2	0.9 ± 0.1	0.5 ± 0.1†	0.9 ± 0.1	0.6 ± 0.1
Osteoid thickness, μm	6.8 ± 0.4	6.6 ± 0.4	6.7 ± 0.3	6.8 ± 0.4
Mineral apposition rate, $\mu m/d$	0.47 ± 0.01	0.44 ± 0.02	0.49 ± 0.02	0.43 ± 0.01†
Mineralizing surface/bone surface, %	3.0 ± 0.4	4.1 ± 0.8†	3.2 ± 0.4	4.1 ± 0.5
Bone formation rate/bone surface, $\mu m^3/\mu m^2 \cdot y$	5.4 ± 0.8	6.7 ± 1.4	5.7 ± 0.8	6.4 ± 0.7
Bone formation rate/bone volume, %/y	12.4 ± 2.0	15.3 ± 3.0	13.2 ± 1.5	15.8 ± 2.1
Trabecular thickness, μm	90.4 ± 3.6	88.4 ± 4.2	85.6 ± 3.4	87.2 ± 4.0
Trabecular number/mm	1.61 ± 0.05	1.58 ± 0.07	1.65 ± 0.06	1.58 ± 0.06

* Data shown are mean ± SE. No differences were seen between placebo and calcitriol groups.
† = $P < 0.02$, initial compared with final (paired t-test).

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the calcitriol did have physiologic effects in our patients. Direct effects of calcitriol on the bone, however, were not observed. Many in-vitro studies have suggested that calcitriol is a potent resorbing agent (2). This was not seen in our in-vivo study in women with postmenopausal osteoporosis. The bone mass did not decrease, the urinary hydroxyproline did not increase, and no increase in resorption was seen histologically. It is possible that calcitriol does not cause increased bone resorption because women with postmenopausal osteoporosis have unresponsive cells. It is also possible that in-vitro resorption will not occur unless there is also low serum calcium or phosphate, both of which were normal in our patients. Studies of normal men given calcitriol suggest that bone resorption occurred only when calcium intake was low (42). Finally, the dose of calcitriol in this study was modest. Higher doses could possibly lead to bone effects, but the oral dose is limited by hypercalciuria or hypercalcemia.

Recently Aloia and colleagues (25) performed a study with a similar design and found that the bone mass remained stable in the calcitriol group but declined in the placebo group. In their study, the definition of hypercalciuria was more than 11.8 mmol/d (400 mg/d). The mean dose was 0.8 μg/d, twice that in our study. Because their patients could tolerate higher doses of the drug, it is possible that there were nutritional or ethnic differences in the patient populations. Their study had only 27 subjects, and the statistical analysis used one-way *t*-tests, so it is also possible that the apparent beneficial results of calcitriol represented a type I statistical error.

Previous reports have suggested that calcitriol may cause renal damage (43). Aloia and colleagues (25) also reported serious nephrotoxicity. It is not certain if this is a direct effect of the drug or a consequence of hypercalciuria or hypercalcemia. In this study, our osteoporotic patients rarely demonstrated an increase in the serum calcium to abnormal levels, even with large increases in urinary calcium. This is in contrast to the experience in uremic patients who frequently develop hypercalcemia. None of our patients had signs of nephrotoxicity, and the mean serum creatinine in the cacitriol group was not different from the placebo group. It is important to note, however, that when we did see a large increase of urinary calcium, we adjusted the dose so that no patient had high urinary calcium sustained longer than 6 weeks. We think there is a potential for serious nephrotoxicity if the serum calcium is only monitored in these patients with osteoporosis.

Calcitriol is not effective in the treatment of established postmenopausal osteoporosis. This hormone may play a role in the pathogenesis of the disease and it may be useful in some patients with osteomalacia, but it is not superior to calcium supplements in women with osteoporosis.

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